

# Lab 3 – Architecture Selection & Justification

## System: Podcast Guest Scheduling & Outline Builder

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### Architecture Selection

We chose **Layered Architecture** for the Podcast Guest Scheduling & Outline Builder system.

#### Reasons for Choosing Layered Architecture

##### 1. Clear separation of responsibilities

The system has different types of work such as handling the user interface, managing bookings, generating the guest's PDF outline, handling calendar notifications, and storing the data. Layered architecture allows these responsibilities to be separated instead of putting every functionality together. For example, the **Booking Management** component can handle booking-related operations while the **Database** is responsible for storing booking and topic information. This makes the system easier to understand and maintain.

##### 2. Suitable for the size and requirements of the system

The system mainly involves a straightforward flow between the guest, booking management, topic submission, calendar service and database. It does not require many independently deployed services. A layered approach keeps the design relatively simple while still allowing the different parts of the system to communicate through defined interfaces such as the **Booking API, Authentication API and Calendar API**.

#### Security Advantage

Layered architecture allows authentication and authorization to be handled separately from the main booking functions. In our design, the **Authentication & Authorization** component controls access before users can access protected information. This is important because the SRS specifies that booking details and guest topics should only be accessible to authorized users.

#### Performance Benefit

The architecture separates the processing responsibilities, so operations such as booking management and PDF generation can be handled by their respective components instead of making every part of the system perform the same work. The **Outline / PDF Generator** can focus specifically on generating the production sheet from submitted topics. This supports the requirement that the formatted PDF production sheet should be generated within **1 second** after topic submission.

